

Autonomous Retail Researcher Agent

Course Name: AGENTIC AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Problem Statement & Objectives

1. Problem Statement

In today's retail ecosystem, organizations rely heavily on market research to understand trends, customer behavior, pricing strategies, and competitive positioning. However, traditional retail research methods are largely manual and involve collecting data from multiple sources such as websites, reports, and articles, followed by analysis and documentation.

This manual process introduces several challenges:

- **Time-Consuming:** Gathering and analyzing data manually requires significant time and effort
- **Inconsistency:** Different analysts may interpret data differently, leading to inconsistent results
- **Information Overload:** The vast amount of online data makes it difficult to extract relevant insights
- **Delayed Decision-Making:** Slow research processes impact business agility
- **Human Bias:** Manual interpretation can introduce subjective bias

Additionally, the dynamic nature of the retail industry demands **real-time insights**, which traditional systems fail to provide efficiently.

Therefore, there is a strong need for an intelligent, automated system that can:

- Collect relevant data in real time
- Analyze it efficiently
- Generate structured and meaningful insights
- Minimize human intervention

2. Project Objectives

The primary objective of this project is to design and develop an **AI-driven autonomous system** that automates the retail research process.

Core Objectives

- **Automation of Research Process**

Eliminate manual data collection and analysis using AI agents

- **Real-Time Data Processing**

Fetch and process up-to-date information from web sources

- **Structured Report Generation**

Convert raw data into meaningful, well-organized insights

- **Multi-Agent Collaboration**

Use specialized AI agents (Researcher and Analyst) for task division

- **Improved Accuracy and Efficiency**

Reduce human error and increase speed of analysis

- **User-Friendly Interaction**

Allow users to input queries in simple natural language

Secondary Objectives

- Ensure system scalability for multiple users
- Maintain modular design for easy upgrades
- Enable integration with future technologies (database, caching, cloud)

3. Scope of the Project

The scope defines what the system **will and will not do**.

In Scope

- Accept user input in the form of a research topic
- Retrieve real-time data using web search APIs
- Process and analyze data using AI agents
- Generate structured reports using LLM
- Display output through a web interface
- Store generated reports temporarily

Out of Scope / Limitations

- No long-term persistent database storage
- No authentication or user account system
- No offline functionality (depends on internet APIs)
- Limited customization of AI-generated reports

Target Users

- Retail analysts
- Business decision-makers
- Students and researchers

1. Key Features

1. Automated Data Collection

The system automatically gathers data from online sources using web search APIs, eliminating the need for manual browsing.

2. AI-Based Data Analysis

The collected data is processed using AI models to extract meaningful insights, trends, and patterns.

3. Multi-Agent Architecture

The system uses a **collaborative AI approach**:

- **Researcher Agent:** Collects raw data
- **Analyst Agent:** Processes and structures data

This division improves efficiency and modularity.

4. Real-Time Insight Generation

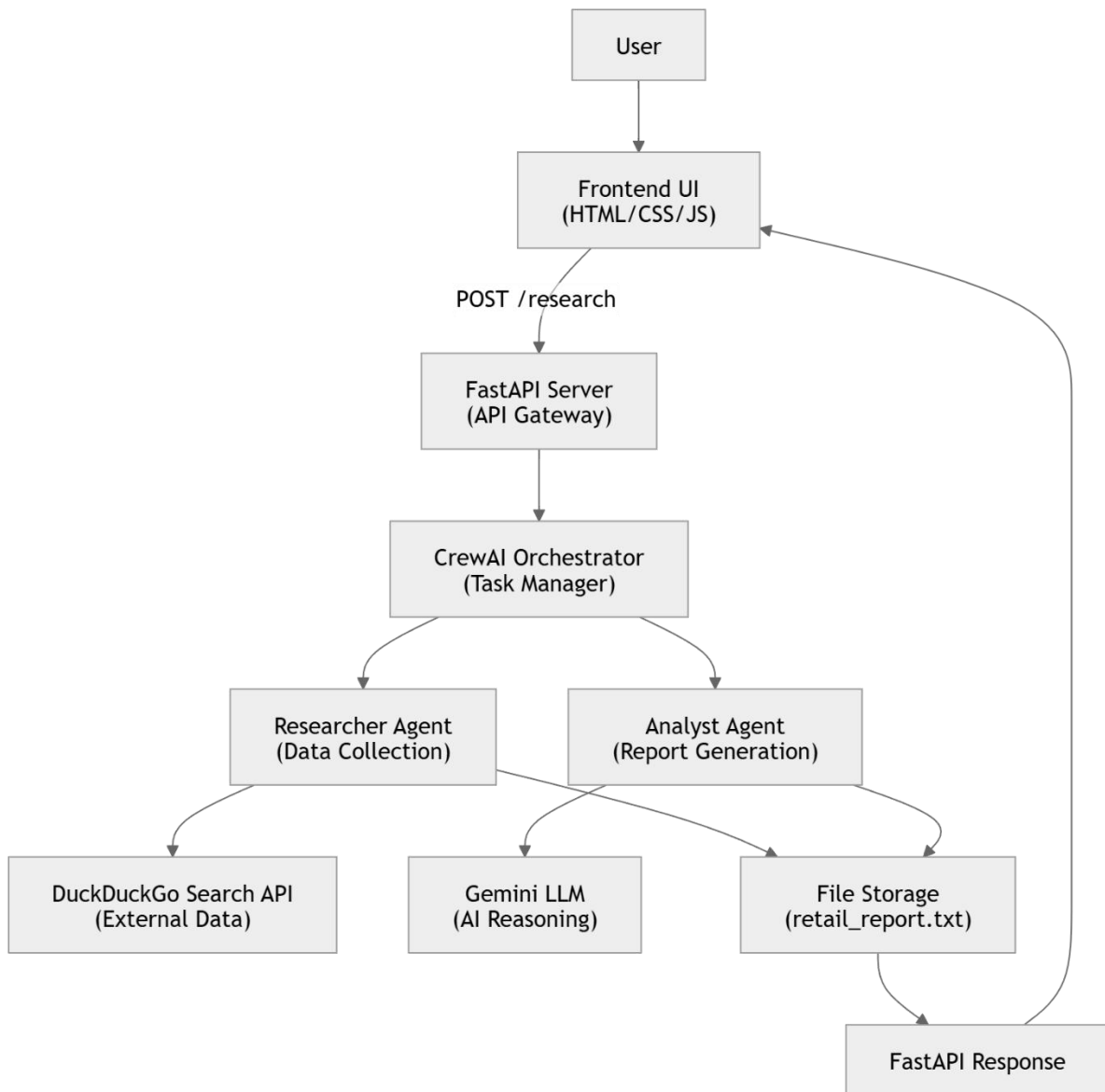
Unlike static systems, the solution provides **up-to-date information**, ensuring relevance.

5. Structured Report Output

The system generates reports in a clear and organized format, making it easy to interpret.

6. User-Friendly Interface

Users can interact with the system using simple text input without requiring technical knowledge.



Detailed Workflow Explanation

Step 1: User Interaction

The user enters a research topic through the frontend interface.

Example: "Impact of AI in retail industry"

Step 2: API Communication

The frontend sends the request to the backend using REST API (FastAPI).
The backend validates the input and ensures proper formatting.

Step 3: Orchestration Engine Activation

The backend triggers the **CrewAI orchestration engine**, which manages task execution among multiple AI agents.

Step 4: Data Collection (Researcher Agent)

The Researcher Agent:

- Uses DuckDuckGo API
- Fetches relevant articles, summaries, and data
- Filters useful information

Step 5: Data Processing (Analyst Agent)

The Analyst Agent:

- Receives collected data
- Structures and refines it
- Identifies key insights

Step 6: Report Generation (LLM)

The processed data is passed to the **Gemini LLM**, which:

- Generates a natural language report
- Ensures coherence and readability

Step 7: Storage and Output

- Sent back to frontend
- Displayed to the user

Architecture Flow Summary

User → Frontend → Backend → CrewAI → Agents → External APIs → LLM → Storage → User

3. Tools & Technologies Used

Frontend Technologies

- HTML – Structure
- CSS – Styling
- JavaScript – Interactivity

Backend Technologies

- FastAPI – API handling
- Python – Core programming

AI Technologies

- CrewAI – Multi-agent orchestration
- Google Gemini – Language model
- DuckDuckGo API – Data retrieval

Storage

- File-based system for storing reports

Results & Output

1. Screenshots / Outputs



Autonomous Retail Researcher

Give the AI a topic. It will search the web, analyze data, and write a CEO-ready report.

e.g., How is AR being used in sneaker retail in 2026?

Generate Report



Autonomous Retail Researcher

Give the AI a topic. It will search the web, analyze data, and write a CEO-ready report.

What retail strategy was followed by Hermes that made it reach where it is today?

Generate Report

 **The Agents are researching the live internet... This usually takes 30-45 seconds.**



Autonomous Retail Researcher

Give the AI a topic. It will search the web, analyze data, and write a CEO-ready report.

What retail strategy was followed by Hermes that made it reach where it is today?

Generate Report

Executive Summary: Hermès' Enduring Retail Strategy for Luxury Market Leadership

Date: October 26, 2023

To: CEO

From: Lead Retail Analyst

Subject: Analysis of Hermès' Strategic Pillars for Sustained Luxury Market Dominance

This summary outlines the foundational retail strategies employed by Hermès that have propelled it to its current esteemed position within the global luxury market. By systematically diverging from conventional retail practices, Hermès has cultivated a unique and highly successful model centered on desire, quality, experience, and exclusivity.

Hermès' Core Strategic Pillars

2. Key Outcomes

- Successfully automated the retail research workflow
- Reduced manual effort and improved efficiency
- Generated accurate and structured insights
- Demonstrated effective multi-agent collaboration
- Enabled real-time data retrieval and processing
- Provided scalable and modular architecture

Conclusion

The Autonomous Retail Researcher Agent demonstrates the practical application of AI in automating complex research tasks.

By integrating multi-agent systems, real-time data retrieval, and large language models, the system provides efficient, accurate, and structured outputs.

The project highlights how AI can significantly improve productivity and decision-making in the retail domain.

Future Scope & Enhancements

1. Database Integration

- Use MongoDB or PostgreSQL
- Store historical reports

2. Caching Implementation

- Use Redis
- Reduce response time

3. User Authentication

- JWT-based login system
- Personalized dashboards

4. Parallel Agent Execution

- Reduce processing time

5. Cloud Deployment

- Deploy on AWS, Azure, or GCP
- Enable scalability

6. Advanced Analytics

- Add charts and visual insights
- Trend prediction features

7. UI/UX Improvements

- Interactive dashboards
- Better visualization